26900370 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCMBG301 - ESSENTIALS OF IMMUNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Notes Questions

Answer any **10** questions

Each question carries **2** marks

1. Which organism is used to produce the smallpox vaccine? [K] / [CO1]
2. What is the antigen-binding site of an immunoglobulin? [K] / [CO2]
3. What is an Arthus reaction? [K] / [CO3]
4. What is acute phase response? [K] / [CO1]
5. Give one example of a tube flocculation test. [K] / [CO2]
6. What is the role of the thymus in the immune system? [K] / [CO1]
7. What is the purpose of using HAT medium in hybridoma technology? [U] / [CO2]
8. What is the role of antibodies in ADCC? [An] / [CO3]
9. What is an endotoxin? [K] / [CO1]
10. Define the complement system. [K] / [CO2]

11. Name the two types of agranulocytes. [K] / [CO1]

12. Name two systemic autoimmune diseases and mention their major target tissues. [K] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. List the secondary lymphoid organs and summarise their functions in the immune response. [K] / [CO1]

14. Compare RIST and RAST based on their purpose and detection. [An] / [CO3]

15. Explain how activated B cells differentiate into antibody-secreting plasma cells. [U] / [CO3]

16. Explain the steps involved in indirect ELISA. [U] / [CO2]

17. What are the major features of acquired immunity? [An] / [CO1]

18. Describe the two stages involved in complement fixation test. [K] / [CO2]

[5x4 = 20]

Part C

Essay Question

Answer any **1** questions

Each question carries **10** marks

19. What are adjuvants, and explain the mechanism of immune stimulation by adjuvants. [U] / [CO2]

20. Explain the primary and secondary immune responses with suitable diagrams. [U] / [CO3]

[10x1 = 10]